Multiple sources of movement-derived resumption: Evidence from Atchan

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Abstract This paper argues for the existence of multiple mechanisms in the grammar that can give rise to resumption in syntactic movement. Empirical motivation for this split comes from extraction patterns in Atchan (Kwa, Côte d'Ivoire), where PP subextraction and subject extraction behave differently: all PP subextraction requires resumption, while subject extraction results in resumption only when a pronoun is extracted. I argue that the first pattern is best captured via a pronunciation requirement or stranding, while cliticization best captures the subject extraction pattern. I thus argue that these resumptive mechanisms are not incompatible, and that they have different empirical coverage.

Keywords: resumption; A'-movement; partial copy deletion; cliticization; stranding

1 Introduction

The motivating question of this paper is what mechanisms force resumptive pronouns to surface in syntactic movement. We know that, across languages, resumptive pronouns can reflect a variety of derivations, for example a repair strategy to improve ungrammatical island-violating derivations (as in English; Ross 1967; Asudeh 2004; Morgan & Wagers 2018, a.m.o.), simple binding of a pronoun with no attempted movement (Cinque 1977; Borer 1984; Aissen 1992; McCloskey 2006), or reduced pronunciation of a lower movement copy (van Urk 2018; Georgi & Amaechi 2023). My interest in this paper focuses on the last derivation: resumptive pronouns arising under syntactic movement.

Of course, not all movement gives rise to resumption. In canonical movement chains, only one copy of the moving element is pronounced:

(1) Who$_i$ did you see { ∅ / *him / *who }$_i$?

In the Copy Theory of Movement (following Chomsky 1993), this single-copy spellout is taken as the default way to spell out a movement chain. Formally, following Nunes (2004); Landau (2006), I assume a Chain Reduction algorithm that ordinarily mandates pronunciation of only one copy in the chain, on grounds of economy (Pesetsky 1998; Landau 2006) or perhaps linearization (Nunes 2004). On this view, the spellout of multiple copies (including the use of resumptive pronouns) is banned, all else being equal, hence the unacceptability of resumption in (1). However, when resumption occurs, we must say something additional to at least partially circumvent the Chain Reduction algorithm.

In the literature, there are multiple mechanisms to circumvent full Chain Reduction and permit resumption. One popular mechanism relies on pronunciation requirements (henceforth P-requirements; Landau 2006; van Urk 2018; Davis et al. 2020; Scott 2021;

Georgi & Amaechi 2023; Yip & Ahenkorah 2023). On this view, certain positions in the clause are associated with prosodic prominence. Lower copies that occupy these prominent positions are insulated from full non-pronunciation, i.e., they cannot be fully deleted by Chain Reduction. In this case, a second algorithm, called Partial Copy Deletion in this paper, shaves off structure from the copy occupying the prominent position, yielding a reduced pronominal element pronounced in that position.

Another line of work proposes that cliticization can circumvent Chain Reduction (Nunes 2004; Kandybowicz 2006; Bošković & Nunes 2007; Harizanov 2014). The intuition in this work is that cliticized elements are invisible to the Chain Reduction algorithm: once a clitic fuses with its host, it no longer sufficiently resembles other elements in the movement chain and hence will not be targeted for deletion. Within this line of research, work is divided on whether this invisibility-causing mechanism should be formalized as DM Fusion (Nunes 2004; Kandybowicz 2006; Bošković & Nunes 2007; following Halle & Marantz 1993) or m-merger (Harizanov 2014; following Matushansky 2006).

Yet another line of work centered in stranding, recently defended by Hewett (2023), proposes that resumptive pronouns in movement-derived resumption do not actually reflect partial realization of lower movement copies. Instead, on this treatment, an apparently resumptive pronoun is generated together with the lower copy of the moving element. The moving element then extracts, stranding the lower pronoun, which is pronounced as an apparent resumptive.

The bodies of work on these mechanisms have only interacted to a limited degree. Work on P-requirements has mentioned, sometimes in passing, that at least some of the resumptive elements those authors focus on are not clitics, rendering the cliticization story irrelevant (cf. van Urk 2018: §2.3; Georgi & Amaechi 2023: Appendix B). Hewett (2023) also raises a number of theoretical and empirical challenges for P-requirement accounts, suggesting instead that stranding provides a cleaner way to capture movement-derived resumptives. Because of this limited degree of interaction, it is not entirely clear whether we actually need to appeal to all of these mechanisms—that is, whether the choice between these various approaches to resumption can be entirely diagnosed on empirical grounds, or whether the different approaches have the same empirical coverage and are largely to be distinguished in their theoretical assumptions.

In this paper, I will argue that multiple of these mechanisms can be active within a single language and that they derive resumption in fundamentally different ways. The novel data in this paper will center on movement-derived resumption in Atchan, a Kwa language of Côte d'Ivoire. I will show that Atchan exhibits splits in movement-derived resumptive morphology along two major axes: the position from which the moving element is extracted (subject vs. PP object), and the kind of moving element (lexical DP vs. pronoun). Specifically, both lexical DPs and pronouns are resumed when extracted from PP object position, but only pronouns are resumed in subject extraction. The particularities of these splits and their morphological realizations support the conclusion that we must appeal to multiple different mechanisms in deriving these resumptive pronouns. Namely, while P-requirements and stranding can both capture PP object resumption, cliticization is best suited to capture pronominal subject resumption. The core argument in this paper is thus that the same properties that make each discussed account of resumption well-suited to capture resumption in one position render that account ill-suited to capture resumption in the other position.

This paper proceeds as follows. In §2, I present background on Atchan and its pronominal system in anticipation of our later discussion of resumption. From there, I introduce Atchan's A'-movement constructions in §3. In §4, I introduce in more detail the approaches to movement-derived resumption under consideration here. In §5, I discuss resumption in PP subextraction; I show that PP subextraction can be captured by a P-requirement or by stranding, but that a cliticization approach would not be well-motivated. Then, in §6, I discuss subject extraction, proposing that resumption in pronominal subject extraction is best captured by cliticization. Finally, §7 concludes.

2 Atchan and its pronominal inventory

2.1 Language and consultant information

Atchan (also known by the exonym Ébrié; ISO: ebr) is a Kwa (Niger-Congo) language spoken by the Tchaman people, who live in villages located in and near the city of Abidjan, Côte d'Ivoire. The city of Abidjan, Côte d'Ivoire's economic capital, was built upon the traditional homelands of the Tchaman people. Recent estimates suggest that there are around 150,000 Tchaman people (Dido 2018). The vast majority of Atchan speakers, including all of my consultants, also speak French, often in addition to other Ivorian languages.

The novel data discussed in this paper comes from elicitation sessions conducted with five speakers of Atchan living in the village of Anono between 2022-2024. Each primary data example includes a reference code corresponding to the date of the elicitation session and the consultant(s) present. Recordings and notes from the Atchan Language Project, including elicitation materials are archived with the California Language Archive (Doko et al. 2021).

Atchan exhibits basic SVO word order. It does not permit pro-drop of animate objects:

(2) Q: Did you see Katie?

he:, mɛ̃= { ŋwu nɛ / ŋwi / #ŋwu }
yes 1SG see 3SG$_{\text{OBV}}$ see.3SG$_{\text{PROX}}$.O see

'Yes, I saw her.' (20240310_kou)

In addition, it does not exhibit morphological case. Observe the identical forms of 'woman' below:

(3) a. **ɓje** é-ɟe cɔ̃
woman PROG-fry fish
'The woman is frying fish.' (20230624_kou_jea)

b. akisí wú **ɓje**
A. see woman
'Akissi saw the woman.' (20230619_kou_jea)

Examples throughout the paper will display a verbal (pre)nasalization process, as exhibited in the contrast between the verbal form beginning in [ŋw] in (2) and [w] in (3b). This process is triggered by the presence of a nasal subject pronoun (like 1SG in (2)). This process and some extensions will be discussed at length in §6.1.2.

<table>
<tr><th></th><th>s position</th><th>o position</th><th>elsewhere</th></tr>
<tr><td>1SG</td><td colspan="3">mɛ̃</td></tr>
<tr><td>2SG</td><td>ɛ</td><td colspan="2">hɛ</td></tr>
<tr><td>3SG$_{\text{PROX}}$</td><td>ã/ɛ̃́∼ń</td><td>-ɛ</td><td>mɛ̃́</td></tr>
<tr><td>3SG$_{\text{OBV}}$</td><td colspan="3">nɛ∼nkɛ</td></tr>
<tr><td>1PL</td><td colspan="3">lo</td></tr>
<tr><td>2PL</td><td>ɔ̃́/hɔ̃́</td><td colspan="2">hɔ̃́</td></tr>
<tr><td>3PL</td><td colspan="3">wo</td></tr>
<tr><td>3INAN</td><td>á/∅</td><td>∅</td><td>ló</td></tr>
</table>

Table 1: Personal pronouns in Atchan following Jarvis 2025c, adapted and updated from Bôle-Richard 1983. Forms separated by / are conditioned by aspect; forms separated by ∼ are dialectal variants.

2.2 Pronominal inventory

Before we can discuss resumption, we must discuss Atchan's pronominal inventory. To this end, Table 1 provides an inventory of Atchan's personal pronouns. Note that the active phi-features in the pronominal system are person, number, and animacy.[1]

Several pronouns take on distinct forms when they occupy canonical subject (S) or canonical object (O) position, i.e., directly adjacent to verbal/auxiliary material. I will introduce the S alternations in more depth in §6.1, when they become relevant. Here, for readability, I note that there are multiple forms in the same cell of several pronouns. The different forms of the 3SG$_{\text{OBV}}$ pronoun, and the 3SG$_{\text{PROX}}$.S forms [ɛ̃́] and [ń], reflect dialect variation, which we can set aside. Meanwhile, the distinction in subject position between 3SG$_{\text{PROX}}$ [ã] and [ɛ̃́], between 2PL [ɔ̃́] and [hɔ̃́], and between 3INAN [á] and [∅], is based on aspect, as will be discussed in more detail later.

One pronoun, 3SG$_{\text{PROX}}$, has a distinct O form, which is conditioned by structural adjacency with the verb. This form, /-ɛ/, phonologically coalesces with the verb, as is illustrated with the following examples:

(4) a. /wú-ɛ/ see-3SG$_{\text{PROX}}$.O → [wî]
  b. /c$^{\text{h}}$rwa-ɛ/ hit-3SG$_{\text{PROX}}$.O → [c$^{\text{h}}$rwɛ]

In the remainder of the paper, I supply the underlying form of the verb for every example of 3SG$_{\text{PROX}}$.O we see.[2]

3 Introducing A'-movement constructions in Atchan

This section presents introductory data on Atchan's A'-movement constructions. As I first argued in Jarvis 2025b; d, relativization and focus/*wh*-movement in Atchan both involve movement.[3] Converging evidence to this effect comes from island sensitivity and crossover effects.

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[1] Atchan has two 3SG animate pronouns, glossed 3SG$_{\text{PROX}}$ and 3SG$_{\text{OBV}}$. As I discussed in Jarvis 2025c, their distributions are constrained by factors like topicality and prominence. This distinction is not crucial for our discussion; the two pattern identically for resumption. I provide both pronominal options when relevant.

[2] In (2), the underlying form of the second provided form was /wu-ɛ/ (with verb's L tone reflecting PFV aspect).

[3] These constructions contrast with matrix topicalization, which does not involve movement (Jarvis 2025a). Unsurprisingly, matrix topics are always 'resumed'.

First, these processes obey standard island constraints on movement, which I, in accordance with Hewett 2023, take as a primary piece of evidence for movement dependencies. For example, they obey the Complex Noun Phrase Constraint (CNPC, Ross 1967). Relative clauses form islands for further relativization (5)[4] and for focus/*wh*-movement (6):

(5) a. mɛ̃= ŋwu [sɛ$_i$ k^{h}ɛ̃́ ∅$_i$ a pɔ ɓje]
1SG see man COMP ASP love woman
'I saw the man who loves the woman.'

b. [ɓje$_j$ k^{h}ɛ̃́ sɛ pɔ ∅$_j$], ɛ̃́= ɲɔ̃
woman COMP man love 3SG$_{\text{PROX}}$.S.PFV pretty
'The woman who the man loves is pretty.'

c. * [ɓje$_j$ k^{h}ɛ̃́ mɛ̃= ŋwu [sɛ$_i$ k^{h}ɛ̃́ ∅$_i$ a pɔ ∅$_j$]], ɛ̃́=
woman COMP 1SG see man COMP ASP love 3SG$_{\text{PROX}}$.S.PFV
ɲɔ̃
pretty
Intended: 'The woman$_j$ such that I saw the man who loves her$_j$ is pretty.'
(20240719_hre)

(6) * t^{h}ábɛ̃$_j$ nɔ̃ ɛ= wu [sɛ$_i$ k^{h}ɛ̃́ ∅$_j$ a pɔ ∅$_i$]
who FOC 2SG.S see man COMP ASP love
Intended: 'Who$_j$ did you see the man that they$_j$ love?' (20240310_kou)

Additionally, these processes in Atchan obey the Coordinate Structure Constraint (CSC), here illustrated with disjunction:

(7) a. * mɛ̃= mpɔ [lep^{h}ã$_i$ k^{h}ɛ̃́ kati wu lizi léka ∅$_i$]
1SG love person COMP K. see L. or
Intended: 'I love the person that Katie saw Lindsay or.' (20240310_kou)

b. * t^{h}ábɛ̃$_i$ nɔ̃ ɛ= wu lizi léka ∅$_i$
who FOC 2SG.S see L. or
Intended: 'Who did you see Lindsay or?' (20240721_hre_jea)

Both of these island-sensitivity observations support the conclusion that these constructions involve syntactic movement.

Additional supporting evidence comes from crossover. These constructions show Strong Crossover (SCO) effects:[5]

(8) a. A.$_j$ nɔ̃ ∅$_j$ a ɗi [lep^{h}ã$_i$ k^{h}ɛ̃́ ɛ̃́$_{*i}$= mpɔ ∅$_i$]
A. FOC ASP COP person COMP 3SG$_{\text{PROX}}$.S.PFV love
'Alexandra is a person$_i$ who she$_{*i}$ loves.'
note: cannot be used to answer the question 'Who loves herself?'

b. t^{h}ábɛ̃$_i$ nɔ̃ ɛ̃́$_{*i}$= mpɔ ∅$_i$
who FOC 3SG$_{\text{PROX}}$.S.PFV love
'Who$_i$ does he$_{*i}$ love?' (20240719_hre)

4 In (5b) and (5c), the relative clause is topicalized due to the consultant's preference in (5b), hence the additional subject pronoun.

5 Note that there is a possible confound in this evidence, since the moved element in each case is fairly local to the subject; it is possible that the ungrammaticality could result from a surface Condition B effect due to the moved element and subject pronoun occurring in the same binding domain. A similar question arises also with the SCO data in (10) and (13) below. Data on SCO in longer-distance phenomena is unavailable at this point.

Again, this is predicted if *wh*-movement/focus and relativization are derived via movement.

In the discussion to come, we will investigate extraction from three structural positions: direct object, PP object, and subject. It is worth confirming that extraction from each position involves movement. This has already been done for direct objects: see the CNPC data in (5), CSC data in (7), and SCO data in (8). We can also confirm the same point for these other structural positions. We also observe systematically here that resumption does not obviate island violations. In the data below, island-violating sentences are constructed to accord with Atchan's general resumption patterns (as discussed further below): i.e., pronominal subjects are resumed, and all PP objects are resumed. Despite this resumption, island effects still obtain.

Subject extraction is sensitive to the CNPC (9), as was previously shown for *wh*-movement in (6), and additionally exhibits Strong Crossover effects (10):[6]

(9) a. * kati$_j$ nɔ̃́ mɛ̃= ŋwu [sɛ$_i$ k^{h}ɛ̃́ ∅$_j$ a pɔ ∅$_i$]
K. FOC 1SG see man COMP ASP love
Intended: 'It's Katie$_j$ who is such that I saw the man who she$_j$ loves.'

b. * mɛ̃́$_j$ nɔ̃́ mɛ̃= ŋwu [sɛ$_i$ k^{h}ɛ̃́ ɛ̃́$_j$= mpɔ ∅$_i$]
3SG$_{PROX}$ FOC 1SG see man COMP 3SG$_{PROX}$.S.PFV love
Intended: 'It's her$_j$ who is such that I saw the man who she$_j$ loves.'
(20240724_hre)

(10) kati nɔ̃́ a ɗi [lep^{h}ã$_i$ k^{h}ɛ̃́ ã$_{*i}$= mú sálé ∅$_i$ e-ɓá gɛ jí]
K. FOC ASP COP person COMP 3SG$_{PROX}$.S think COMP PROG-FUT win thing
'Katie is a person$_i$ who she$_{*i}$ thinks will win.' (20240819_hre)

As we will see extensively later, lexical DPs and pronouns have different morphological reflexes in subject extraction, but both display identical CNPC sensitivity, as shown in (9).

Subextraction of PP objects, too, obeys these same constraints: CNPC data is shown in (11), CSC data in (12), and SCO data in (13):

(11) * mɛ̃= ŋwu [ájá$_j$ k^{h}ɛ̃́ mɛ̃= mpɔ [jípɔ́mã$_i$ k^{h}ɛ̃́ ∅$_i$ a hrɔmã ló$_j$ mɑ̃́ɲɟi]]
1SG see tree COMP 1SG love woman COMP ASP hide 3INAN behind
Intended: 'I saw the tree$_i$ such that I love the woman who hid behind it$_i$.'
(20240423_kou)

(12) * mɛ̃= ŋwu [ájá$_i$ k^{h}ɛ̃́ kati hrɔmã [tábrɛ t^{h}é léka ló$_i$ mɑ̃́ɲɟi]]
1SG see tree COMP K. hide table under or 3INAN behind
Intended: 'I saw the tree$_i$ such that Katie hid under a table or behind it$_i$.
(20240521_kou)

(13) aleksandra nɔ̃ a ɗi [lep^{h}ã$_i$ k^{h}ɛ̃́ ɛ̃́$_{*i}$= nɛ̃́ mɛ̃́$_i$ mɑ̃́ɲɟi]
A. FOC ASP COP person COMP 3SG$_{PROX}$.S.PFV look 3SG$_{PROX}$ behind
'Alexandra is a person$_i$ who she$_{*i}$ looked behind.' (20240801_hre)

[6] I do not discuss CSC effects, since this paper focuses on subjects which are both linearly- and structurally-adjacent to the verb. Note also that a parallel to (10) with pronominal extraction cannot be tested for SCO, due to obligatory resumption of pronominal subjects. That is, in (10), we force SCO-violating extraction of 'person' from the lower subject position through the gap. In contrast, due to obligatory resumption in pronominal subject extraction, there is always an alternate derivation available involving local extraction from the higher position.

Note that resumption systematically fails to rescue CNPC and CSC island violations here, just as in (9b). Note that, in §5, we will see some examples of relativization with additional dislocation of the subject to guarantee a raising derivation (cf. Jarvis 2025b); following that work, a raising derivation is also available for relativization without subject dislocation, so the ungrammaticality of this data speaks in favor of both raising and non-raising derivations involving movement. To my knowledge, subject dislocation similarly has no effect on the other movement diagnostics.

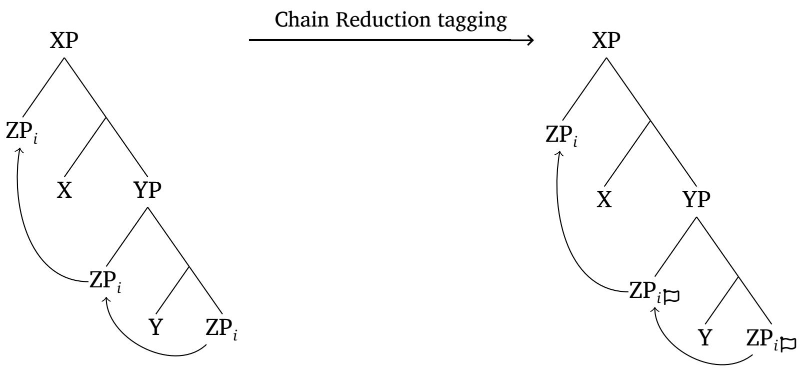
With this, I conclude that relativization and *wh-*/focus movement in Atchan necessarily involve movement in all three structural positions under discussion in this paper. Our goal for the rest this paper is to assess which accounts of movement-derived resumption can plausibly capture the resumption patterns that we observe in Atchan A'-movement. To this end, the next section provides a short overview of the three accounts under consideration in this paper. After that discussion, we will assess how well those accounts can capture different facets of the resumption patterns to be observed in Atchan.

4 Introducing three accounts of movement-derived resumption

In this section, I introduce the three accounts of movement-derived resumption that we will consider in this paper: P-requirements, cliticization, and stranding.

All three accounts take as background the Copy Theory of Movement (following Chomsky 1993), i.e., that syntactic movement leaves behind full copies of the movement element. To explain non-resumption, all of these accounts assume a *Chain Reduction* algorithm, which mandates that lower copies in a movement chain go unpronounced. This treatment of Chain Reduction as an algorithm comes from the work of Nunes (2004) (though Nunes's Chain Reduction algorithm crucially references the linearization algorithm, which is not the focus of this work). More recent work on movement-chain resolution (e.g., Scott 2021; Mendes & Ranero 2021) adopts this term for the relevant algorithm without an appeal to linearization.

For presentational purposes, I will treat Chain Reduction as a postsyntactic algorithm which *tags* lower movement copies (i.e., copies which are c-commanded by a higher movement copy) for non-pronunciation at PF. Schematically, I will represent the result of Chain Reduction as a non-pronunciation tag ⚐ applied to each lower movement copy, as is illustrated below for movement of an element ZP:

(14)


In standard circumstances, the non-pronunciation tag will lead to deletion of each lower movement copy before PF. Note that, in much work that involves the Chain Reduction algorithm, Chain Reduction is taken to itself cause deletion (rather than to lead to later deletion via tagging); I couch the overview of the analyses here in a tagging treatment for presentational simplicity, though this is not crucial.

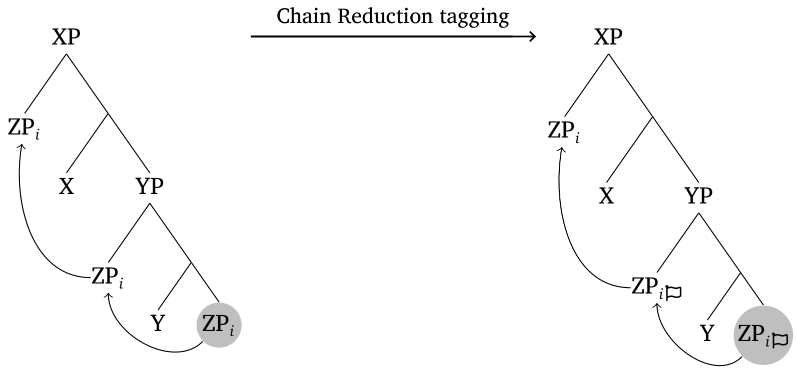
Each of the accounts of movement-derived resumption under discussion in this paper provides a way for this tagging-and-deletion process to be disrupted in some way. I introduce P-requirement approaches in §4.1, cliticization approaches in §4.2, and stranding approaches in §4.3.

4.1 P-requirement approaches to movement-derived resumption

In a nutshell, analyses centered in P-requirements, following Landau (2006), suggest that certain positions resist total deletion after Chain Reduction tagging; this line of work has been furthered and applied to resumptive pronouns by van Urk (2018); Scott (2021); Georgi & Amaechi (2023); Yip & Ahenkorah (2023), a.o.

As discussed previously, each lower copy tagged by Chain Reduction will ordinarily be deleted at PF. However, the account developed by Landau (2006) holds that certain positions in a movement chain can be associated with a pronunciation (P-)requirement and therefore incompatible with wholesale deletion. In this discussion, I will schematize positions associated with P-requirements with a shaded circle. For example, if the Comp,Y position in the tree in (15) is associated with a P-requirement, we would arrive at the representation shown below:

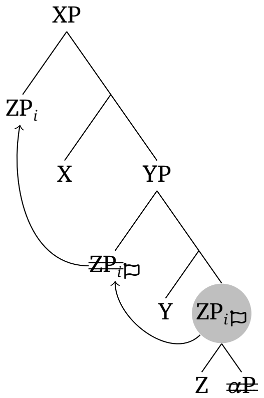
(15)



This gives rise to a tension: the P-requirement calls for pronunciation of material in the Comp,Y position, while the deletion tag pushes towards non-pronunciation.

P-requirement accounts posit that this tension is resolved by realizing only a portion of the offending copy's syntactic structure, with an additional algorithm determining the amount of structure that is realized. Following van Urk (2018) and Georgi & Amaechi (2023), I refer to this additional algorithm as *Partial Copy Deletion*. Schematically, this algorithm deletes some portion of the lower movement copy occupying the prominent position; for example, a smaller portion αP within ZP might be deleted, with just the head Z spelled out:

(16) **Partial Copy Deletion of Comp,Y copy of ZP**



Deletion here is indicated through double strikeout. This interaction between P-requirements and Chain Reduction/Partial Copy Deletion thus derives total non-pronunciation in positions not subject to P-requirements (as in the copy of ZP occupying Spec,YP) as well as resumptive spellout in positions subject to P-requirements (as in the Comp,Y copy).

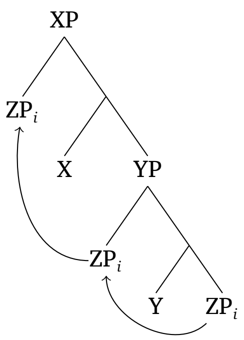
Work that applies P-requirements to derive resumptive pronouns (van Urk 2018; Scott 2021; Georgi & Amaechi 2023; Yip & Ahenkorah 2023) generally assumes an articulated DP structure, in which pronouns and lexical DPs both contain multiple internal projections (and, often, lexical DPs contain certain additional projections that pronouns lack). See van Urk 2018 and Georgi & Amaechi 2023 for relevant discussion.

4.2 Cliticization approaches to movement-derived resumption

The central idea of cliticization-based analyses of resumptive pronouns (Harizanov 2014; Kramer 2014) is that cliticized elements do not 'look like' other copies in the movement chain. That is, by virtue of undergoing cliticization, clitics are able to 'hide from' the Chain Reduction algorithm. Since Chain Reduction does not 'see' these clitic elements, the algorithm does not tag them for deletion; as a result, the clitic is exponed as though it were never part of a movement chain.

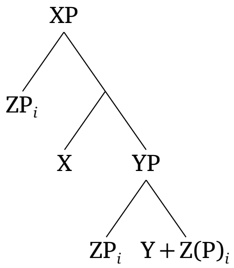
This derivation involves a series of four ordered steps, which I schematize here. In the first step, an element (here, ZP) moves; at this point, the element destined to be a clitic leaves copies in lower intermediate movement sites, as is standard. The tree below shows ZP moving from Comp,Y through Spec,YP on its way to Spec,XP.

(17) **Step 1: Movement of ZP**



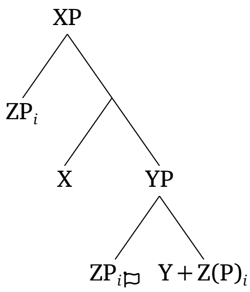
In the second step, post-syntactic cliticization occurs. This cliticization operation is often formalized as DM Fusion (Nunes 2004; Kandybowicz 2006; Bošković & Nunes 2007; cf. Halle & Marantz 1993; 1994 on the Fusion operation), which I will assume in this presentation of the cliticization view and return to in §6.6.1. For this presentation, we will assume that cliticization affects the Comp,Y copy (though, of course, this account could apply equally well[7] to other copies). The result of this Fusion is a combined feature bundle containing the features of both Y and Z(P).[8]

(18) **Step 2: Fusion of Y and ZP**



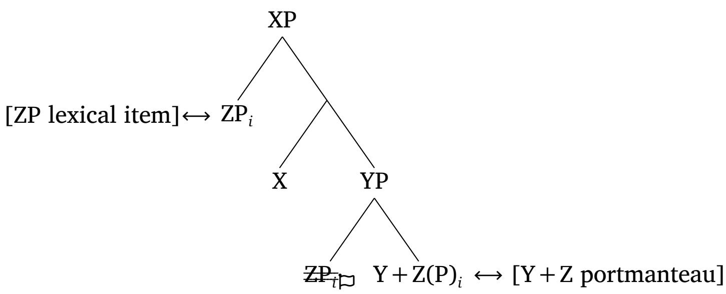
In the third step, the Chain Reduction algorithm applies; it leaves the highest copy of ZP intact and tags all other *visible* copies in the ZP movement chain for non-pronunciation (tag illustrated with ⚐). Crucially, the Fused copy is not visible to Chain Reduction, so that copy remains untagged.

(19) **Step 3: Chain Reduction tags visible copies for non-pronunciation**



Finally, the fourth step involves Vocabulary Insertion. Here, the highest copy is pronounced in full, and tagged copies go unpronounced. A portmanteau form is inserted to realize the Fused Z(P) + Y feature bundle.

(20) **Step 4: Vocabulary Insertion**



Overall, we see that cliticization of Z to Y leads to double exponence of Z's features: the highest copy of ZP is spelled out as the head of the movement chain, and the Fused copy is spelled out as part of a portmanteau. In this way, cliticization is provides a route

[7] In fact, most work that appeals to cliticization assumes that cliticization targets a copy occupying a specifier position. Here, I show cliticization impacting the lowest position for parallelism in presentation across all three accounts.

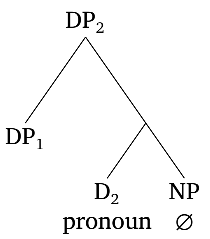
[8] For the purposes of this illustration, I do not focus on the distinction between heads and phrases in Fusion; see Harizanov 2014 and Kramer 2014 for discussion of DM operations similar to this one applying to branching phrasal projections.

to resumption that does not need to appeal to the Partial Copy Deletion algorithm: since the clitic copy is never tagged for deletion, the Partial Copy Deletion algorithm is never invoked.

4.3 Stranding approaches to movement-derived resumption

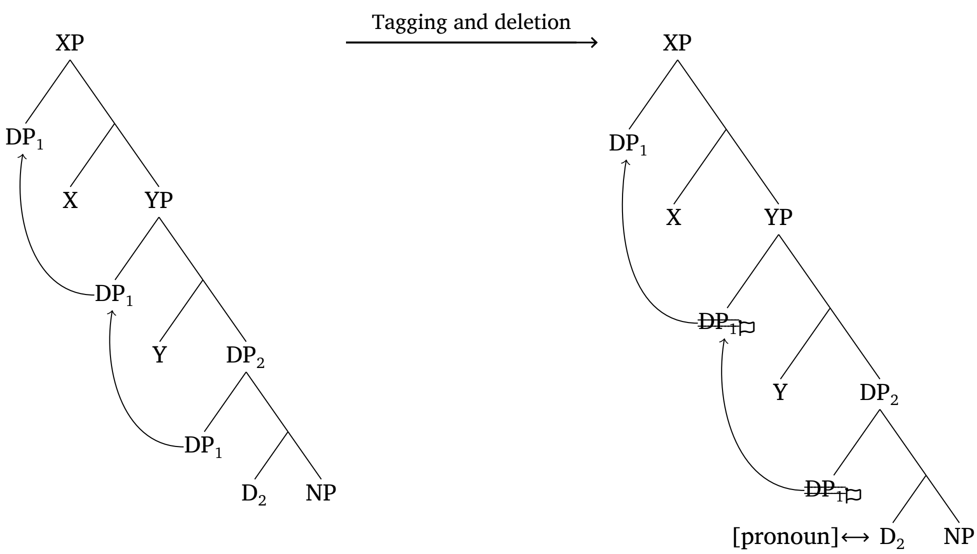
Both P-requirement accounts and cliticization accounts provide a mechanism to generate resumptive pronouns by realizing lower movement copies. In contrast, accounts based in stranding suggest that lower movement sites are more complex than we have assumed to this point, providing a way to derive movement-derived resumptive pronouns that do not technically realize lower movement copies. Here, I largely present the implementation of the stranding analysis in Hewett 2023.

Hewett's implementation of the stranding approach draws on Big DP approaches to clitics (Uriagereka 1995; Torrego 1996, a.m.o.).[9] On Hewett's analysis, the base position of a moving element DP_1 can be more complex than we have assumed; namely, it can be generated inside a pronominal DP_2:

(21)


This account assumes that the NP here is phonologically null and that D_2 is realized as a pronoun (cf. Elbourne 2005, a.o.).

From here, DP_1 can subextract from within this larger structure, as shown in the left-hand tree in (22) below. The result of this subextraction is a movement chain that only involves DP_1. In Chain Reduction, all lower and intermediate copies of DP_1 will be deleted, as shown in the right-hand tree in (22). Since D_2 is not related to DP_1 by a movement chain, it is not deleted and instead is realized as a pronoun.

(22)


9 For terminological clarity, note that I refer to the approach sketched in this section as *stranding*, to be distinguished from the *cliticization* account in §4.2.

In this way, the stranding approach derives resumptive pronouns that, while occurring in sentences with syntactic movement, are not the spellout of lower movement copies. On this view, all lower copies of the moving element DP$_1$ are fully deleted after Chain Reduction. Instead, the lower pronoun that we see is the realization of additional structure that surrounds the lowest movement copy.

4.4 Movement-derived resumption: Outlook

In the next two sections, we will focus on resumption in Atchan movement constructions (focus/*wh*-movement and relativization). Our major generalization will be that whether a moving item is resumed depends both on the identity of the moving item (lexical DP or pronoun) and where the item is moving from. These patterns are summarized in Table 2.

	DO	PP object	Subject
Pronoun extraction	∅	✓	✓
Lexical DP extraction	∅	✓	∅

Table 2: Summary of resumption patterns in Atchan movement phenomena.

It is important to note that not all extraction in Atchan results in resumption. Notably, when a direct object is extracted, it cannot be resumed. This holds across A'-movement phenomena (focus/*wh*, relativization) and across DP types (lexical DP, pronoun):[10]

(23) a. katí nɔ̃ mɛ̃= { ŋwu / *ŋwu nɛ / *ŋwi }
K. FOC 1SG see see 3SG$_{\text{OBV}}$ see.3SG$_{\text{PROX}}$.O
'It's Katie that I saw.' (20240310_kou)

b. mɛ̃́ nɔ̃ mɛ̃= { ŋwu / *ŋwu nɛ / *ŋwi }
3SG$_{\text{PROX}}$ FOC 1SG see see 3SG$_{\text{OBV}}$ see.3SG$_{\text{PROX}}$.O
'It's her that I saw.' (20240310_kou)

c. mɛ̃= mpɔ [lep^{h}ã k^{h}ɛ̃́ ɛ= { wu / *wu nɛ / *wi }]
1SG love person REL 2SG.S see see 3SG$_{\text{OBV}}$ see.3SG$_{\text{PROX}}$.O
'I love the person who you saw.' (20240310_kou)

d. katí nɔ̃ mɛ̃= mú sálé mɛ̃= { ŋwu / *ŋwu nɛ / *ŋwi }
K. FOC 1SG think COMP 1SG see see 3SG$_{\text{OBV}}$ see.3SG$_{\text{PROX}}$.O
'It's Katie who I think that I saw.' (20240719_hre)

Neither pronouns (23b) nor lexical DPs (23a, 23c) can be resumed. The same occurs in long-distance extraction of lexical DPs (23d); to my knowledge, long-distance extraction of pronouns is parallel. Since Atchan does not permit animate object pro-drop, the only way to derive non-resumption in (23) is to genuinely not pronounce the lower movement copy. This provides language-internal evidence that Chain Reduction is active in Atchan.

As can be seen in Table 2, however, the pattern is different in extraction from other syntactic positions. The next two sections will focus, respectively, on PP subextraction and subject extraction. I will argue that the three accounts discussed in this section are each able to capture only part of the data. In §5, I will show that resumption in PP subextraction is well-captured on P-requirement and stranding approaches, while cliticization

[10] The 3SG$_{\text{PROX}}$ possibilities shown here involve the verb form [(ŋ)wi], underlyingly /wu-ɛ/.

accounts are difficult to motivate here. In contrast, §6 will argue that the subject extraction data are best captured on a cliticization account, while P-requirement and stranding approaches struggle to capture this data. Subject pronouns, I will argue, are syntactic clitics, but subject lexical DPs are not, providing an independently-motivated explanation for the subject resumption split.

5 PP object extraction: Across-the-board resumption

In this section, we focus on the extraction of postpositional objects in Atchan.[11] The resumption patterns are introduced in §5.1, followed by consideration of possible analyses.

5.1 Empirical data

When an item is subextracted from a PP, it must be resumed by an element (from the 'elsewhere' pronoun series) that matches it in all phi-features.

This is shown with relativization[12] below: we see an inanimate relative-clause head (24b), a singular animate head (24c), and a plural animate one (24d). Each of these relatives involves subject dislocation (of 'Katie'), which I showed in Jarvis 2025b forces a raising relative-clause structure. This means that the element that moves in these relative clauses is the lexical nominal head. We therefore can conclude that the resumptive pronoun we see in each example is a reduced lower movement copy.

(24) a. kati hrɔmã [ájá/ɓje/ɛ̃́mjé/hɔ̃́ mɑ̃́nɟi]
K. hide tree/woman/women/2PL behind
'Katie hid behind the tree/woman/women/you (pl.).'

b. kati$_j$ já$_i$ k^{h}ɛ̃́ ∅$_j$ a hrɔmã { ló / *mɛ̃́ / *ájá / *∅ }$_i$ mɑ̃́nɟi
K. tree COMP ASP hide 3INAN 3SG$_{PROX}$ tree behind
'the tree that Katie hid behind'

c. kati$_j$ ɓje$_i$ k^{h}ɛ̃́ ∅$_j$ a hrɔmã { mɛ̃́ / *ló / *ɓje / *∅ }$_i$ mɑ̃́nɟi
K. woman COMP ASP hide 3SG$_{PROX}$ 3INAN woman behind
'the woman that Katie hid behind'

d. kati$_j$ mje$_i$ k^{h}ɛ̃́ ∅$_j$ a hrɔmã { wo / *mɛ̃́ / *ló / *ɛ̃́mje / *∅ }$_i$ mɑ̃́nɟi
K. woman.PL COMP ASP hide 3PL 3SG$_{PROX}$ 3INAN woman.PL behind
'the women that Katie hid behind' (20241217_kou)

Comparing these examples, we see that the resumptive pronouns here must match their antecedents in animacy and number. Note that non-resumption is ungrammatical across the board, as is full lower-copy pronunciation of the lexical DP.

To investigate person-matching effects, we can look at pronominal relativization. In (25), we see obligatory matching of person in addition to number and animacy:

[11] As in many languages, it is difficult in Atchan to determine whether Atchan 'postpositions' are true postpositions of category P or possessed relational nouns. Extracted 'ordinary' possessors and postpositional objects are resumed identically in Atchan, so this distinction is not crucial to us.

[12] The examples in this section focus on relativization. Speakers disprefer subextraction in focus and *wh*-movement. I suspect that this is because pied-piping is permitted in Atchan focus and *wh*-movement but not in relativization.

(25) hɔ̃́ kʰɛ̃́ mɛ̃= hrɔmã { hɔ̃́ / *hɛ / *wo / *mɛ̃́ / *ló / *∅ } mɑ̃́ɲɟi
2PL COMP 1SG hide 2PL 2SG 3PL $3SG_{PROX}$ 3INAN behind
'you (pl.) who I hid behind' (20240423_kou)

In Jarvis 2025b, I showed that some Atchan relatives do not support subject dislocation; this includes the pronominally-headed relatives shown here. It is therefore not entirely possible to diagnose whether (25) has a raising structure or a head-external one. Regardless, I conclude that, insofar as it is testable, person patterns like other phi-features in Atchan resumption: resumption, when it occurs, is fully phi-matching.

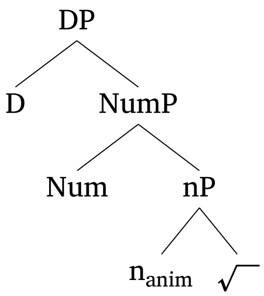
Let us consider how well each of the accounts of movement-derived resumption discussed in the previous section can capture this data. In §5.2, I show that a P-requirement analysis can capture this data fairly straightforwardly. In §5.3, I show that the same is true of a stranding account. In §5.4, I show that a cliticization analysis is difficult to motivate for this data, largely on the grounds that the resumptive pronouns here do not display any traits of clitics.

5.2 P-requirement analyses

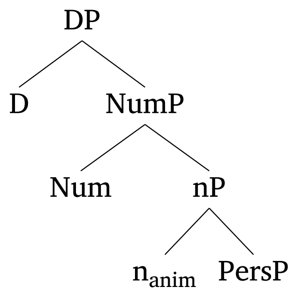
P-requirement analyses can straightforwardly capture this PP object resumption by positing that the complement of P is a position subject to a P-requirement. As illustrated in §4.1, the lower copy of the moving element in Comp,P is tagged for deletion but occupies a prominent position, giving rise to partial deletion and hence resumption in that position. In fact, the complement of P in other languages has also been suggested to be subject to a P-requirement, as in Scott (2021) (Swahili) and Georgi & Amaechi (2023) (Igbo).

P-requirement analyses standardly assume that lexical DPs and pronouns have similar structure. An example set of structures from Scott 2021: 16 is illustrated below (note that, for Scott, n hosts animacy features):

(26) a. Lexical DP structure



b. Pronoun structure



These accounts standardly assume that, in one way or another, the cluster of phi-features present in the nominal structure maps to a pronoun when no root projection is present (as in Elbourne 2005). An example set of Vocabulary Items for Atchan elsewhere pronouns is provided below, in a Scott 2021-style treatment:

(27) a. [D] ⟷ /ló/ (3INAN)
b. [D + sg + anim] ⟷ /mɛ̃́/ ($3SG_{PROX}$)
c. [D + pl + anim] ⟷ /wo/ (3PL)
d. [D + pl + anim + 2] ⟷ /hɔ̃́/ (2PL)

These VIs can realize the structure in (26b), with the standard assumption that the most specified VI is inserted to realize the relevant structure. I assume that, when a root is present in (26a), other non-pronominal determiners are inserted instead.

Now, turning to resumption, we must elaborate how the Partial Copy Deletion applies in Atchan. The obvious approach is to posit that Partial Copy Deletion deletes only the root, leaving all other projections intact. For pronouns, this deletion is vacuous; all structure in (26b) is preserved, so we predict fully phi-matching resumptive pronouns. In the lexical-DP structure in (26a), once the root is deleted, the remaining features will be realized by the best-matching form in (27a–27c). This successfully predicts that the resumptive forms we see in Atchan are (unsurprisingly) always pronouns, and that these pronouns match their antecedents in all phi-features.

I briefly note that the Atchan data call for some refinements to existing cross-linguistic implementations of the Partial Copy Deletion algorithm. The system in van Urk 2018 tries to predict the existence of fully phi-matching resumption via deletion of the nP phase. This derives number and person-matching, but van Urk does not address animacy. If animacy is, as Scott assumes, hosted on n, van Urk's typology must be expanded to allow for deletion only of the root.

While van Urk's system is intended to predict fully phi-matching resumption, the account of Georgi & Amaechi (2023: 994) specifically predicts that fully phi-matching resumption is impossible; they propose that Partial Copy Deletion involves obligatory deletion of the lowest phrasal projection within a nominal's internal structure. Accordingly, as they develop it, the system cannot capture the Atchan data (since, for example, they predict that at least PersP in the pronominal structure must be deleted). A possible repair would be to assume head movement inside the nominal, evacuating the lowest phrasal projection (as in Martinović 2024, cf. Georgi & Amaechi 2023: 992), though no independent evidence for such an analysis in Atchan is available at this point. Note that Georgi & Amaechi, like van Urk, do not address animacy, and more refinement of their account would be needed to incorporate animacy distinctions.

One major question that arises for P-requirement analyses is what motivates that requirement in the relevant position. Sometimes, a clear phonological motivation can be posited (as in Scott 2021 on Swahili, whose explanation centers on wordhood minimality requirements), but other times, the motivation is much less clear (as in Georgi & Amaechi 2023 on Igbo). Since the Atchan resumptive pronouns observed here are phonologically large and separable, it is clear that we cannot appeal to a wordhood minimality explanation like Scott's. There is a limited amount of evidence in Atchan that the left edges of certain phrases require additional morphological material (see Jarvis 2025c: (62) for relevant discussion). However, at this point there is limited PP-specific evidence for prominence in Atchan. Instead, it may be the case that the Atchan P-requirement is best analyzed, along the lines of van Urk 2018, as a requirement that Vocabulary Insertion be run in PP object position; on this view, there may be no phonological motivation as such for the P-requirement.

5.3 Stranding analyses

A stranding approach can also capture this data in a relatively straightforward manner. We must simply claim that, in PP object position, elements that will be extracted (i.e., elements that bear A'-features like [REL]) are generated inside a Big DP structure. As in §4.3, the moving element then subextracts, with the Big DP structure resulting in a resumptive pronoun.

Note that, on this treatment, it would be expected that the resumptive pronouns here match their antecedents in all phi-features. That is, we generally need a mechanism to enforce phi-matching in clitic doubling languages like Spanish. Hewett 2023: 186

suggests a mechanism based on Spec-Head Agree; the mechanism could work the same way in Atchan.

A broader question that arises on any stranding analysis of obligatory resumption, which I cannot fully address here, is obligatoriness of resumption: why is the Big DP structure by hypothesis obligatory in PP object position but not in other positions (like direct object position)? Since Atchan does not exhibit widespread pro-drop (i.e., D_2 cannot usually be null), it is important to have some explanation of when the Big DP structure can occur. For now, this must be left to future research.

5.4 Cliticization analyses

In contrast to P-requirements and stranding, the cliticization analysis discussed in §4.2 is not able to straightforwardly capture this data. On the cliticization analysis, resumptive forms arise when a low movement copy cliticizes to a host; this kind of account is, of course, most compelling when we can find evidence of this cliticization. When no such evidence is available, as noted by van Urk (2018) for Dinka and Georgi & Amaechi (2023) for Igbo, a cliticization analysis is rendered much less attractive.

In Atchan PP subextraction, the resumptive forms that we see are phonologically heavy and separable and do not resemble clitics, either phonologically or syntactically. The pronominal forms that surface here are Atchan's 'elsewhere' set of pronouns; phonologically, these 'elsewhere' forms are all [CV], meeting Atchan's wordhood minimality requirement (Russell 2024). Syntactically, these are the forms that surface in left-peripheral positions (28a) [repeated from (23b) with modified emphasis], can be separated from their following postposition by focus-sensitive particles like the additive *lɔ* 'too' (28b), and can be coordinated (28c):

(28) a. **mɛ̃́** nɔ̃ mɛ̃= ŋwu
3SG$_{\text{PROX}}$ FOC 1SG see
'It's her that I saw.' (20240310_kou)

b. mɛ̃= hrɔmã [**mɛ̃́** lɔ mɑ̃́ɲɟi]
1SG hide 3SG$_{\text{PROX}}$ too behind
'I hid behind her too.' (20240721_hre_jea)

c. mɛ̃= nc^{h}rwa [**mɛ̃́** léka nɛ]
1SG hit 3SG$_{\text{PROX}}$ or 3SG$_{\text{OBV}}$
'I hit either him or her.' (20240724_hre)

This evidence suggests that a cliticization-based analysis of Atchan PP subextraction resumption would not be well-founded.

6 Subject extraction: Split resumption

In this section, I turn to discuss resumption of extracted subjects. Here, I will argue, we should appeal to different analytical tools. In this position, we observe a split between pronoun extraction and lexical DP extraction: pronouns are resumed but lexical DPs are not. I will link this behavior to the observation that pronominal subjects do indeed display a range of clitic-like properties, in contrast to lexical DPs and the 'elsewhere' pronominal forms. I argue that analyses of subject resumption based on P-requirements or stranding are not very tenable; instead, cliticization provides a more promising analysis of this data.

	S form	elsewhere
1SG	—	mɛ̃
2SG	ɛ	hɛ
3SG$_{PROX}$	ã/ɛ̃́	mɛ̃́
3SG$_{OBV}$	—	nɛ
1PL	—	lo
2PL	ɔ̃́/hɔ̃́	hɔ̃́
3PL	—	wo
3INAN	á/∅	ló

Table 3: Elsewhere and S pronouns in Atchan. The absence of a morphologically distinct S form is indicated with '—'.

This section begins by presenting background on Atchan subject pronouns in §6.1. From there, I present the core subject resumption data in §6.2. In §6.3, I argue against using a P-requirement account to derive this pattern; in §6.4, I suggest that similar difficulties also arise on a stranding-based account. Instead, I sketch a cliticization-based analysis in §6.5. Finally, §6.6 discusses loose ends and questions for future work.

6.1 Subject pronouns background

In this section, I discuss background on the syntax and forms of Atchan subject pronouns. The overall argument of this section, in accordance with Russell (2024), is that *all* Atchan subject pronouns pattern differently from subject lexical DPs, in that pronouns are more closely connected with verbal and aspectual material than lexical DPs are; and furthermore that singular subject pronouns are yet more closely connected with verbal/aspectual material than plural subject pronouns are. That is, the picture that emerges is one on which subject pronouns have traits of phonological and syntactic clitics. I begin by discussing subject-specific S pronoun forms in §6.1.1 and then discuss subject pronouns more broadly in §6.1.2.

6.1.1 Dedicated S pronoun forms

Several Atchan pronouns—2SG, 3SG$_{PROX}$, 2PL, and 3INAN—have designated forms that I term S ('subject') forms. These forms are indicated in Table 3, a modified version of Table 1. Note that all of these S forms have a [V] shape. Russell (2024) argues that Atchan's wordhood minimality requirement is [CV] and that these reduced S forms are, therefore, phonological clitics.

These S forms surface only when the pronoun occupies canonical subject position, i.e., when it is adjacent with the verb or preverbal auxiliaries/negation. Examples of 3SG$_{PROX}$.S forms before verbs and auxiliaries are shown below:

(29) 3SG$_{PROX}$.S pronoun forms

a. ã= mpʰí ɗja
3SG$_{PROX}$.S laugh V.PART
'She is laughing.' (20240720_hre_jea)

b. ã= mɑ̃́ mpʰí ɗja
3SG$_{PROX}$.S FUT laugh V.PART
'She will laugh.' (20240807_hre)

c. ɛ̃́= mpʰí ɗja ɛ̃mpi
3SG$_{\text{PROX}}$.S.PFV laugh V.PART one.day.away
'She laughed yesterday.' (20240720_hre_jea)

In contrast, if the subject pronoun is separated from the verb/auxiliary, e.g. by a focus-sensitive particle or a relative clause, the S form is illicit. Below, we see the elsewhere 3SG$_{\text{PROX}}$ form [mɛ̃́]:

(30) 3SG$_{\text{PROX}}$ elsewhere pronoun forms

a. [{ mɛ̃́ / *ã= } brɛ] ã= mpʰí ɗja
3SG$_{\text{PROX}}$ 3SG$_{\text{PROX}}$.S only 3SG$_{\text{PROX}}$.S laugh V.PART
'Only she is laughing.' (20240724_hre_jea)

b. { mɛ̃́ / *ã= } kʰɛ̃́ ã= mpʰí ɗja
3SG$_{\text{PROX}}$ 3SG$_{\text{PROX}}$.S COMP 3SG$_{\text{PROX}}$.S laugh V.PART
'she who is laughing' (20240806_kou_jea)

This provides evidence of these S forms' syntactic weakness (cf. the discussion of weak pronouns in Cardinaletti & Starke 1999, a.m.o.). Note also that these S forms cannot straightforwardly be analyzed as mere exponents of nominative case, since they are conditioned by direct adjacency with the verb/auxiliary.

The distribution and nature of S forms also specifically support a view on which subject pronouns are tightly connected with aspect. Most obviously, subject pronouns *can* form aspectual portmanteaux, but lexical DPs cannot. The alternations between the multiple cells in the S column of Table 3 are all conditioned by aspect.[13] For 3SG$_{\text{PROX}}$, [ɛ̃́] occurs in perfective aspect, and [ã] in other aspects; for 2PL, [ɔ̃́] occurs in non-progressive aspect, with the elsewhere form [hɔ̃́] in progressive aspect; and for 3INAN, [á] occurs in non-progressive aspect, and [∅] in progressive aspect. The 3SG$_{\text{PROX}}$ alternation was shown above in (29). The similar 2PL contrast is shown below:

(31) a. hɔ̃́= e-pʰí ɗja
2PL PROG-laugh V.PART
'You (pl.) are laughing.'

b. ɔ̃́= mpʰí ɗja
2PL.S laugh V.PART
'You (pl.) laughed.' (20240719_hre)

In contrast, lexical DPs in Atchan do not form aspectual portmanteaux.[14]

[13] The analysis of these S forms as pronouns does not accord with Nevins's (2011) claim that pronouns are necessarily TAM-invariant. Note, however, that if Atchan subject pronouns were analyzed as agreement, this would constitute an instance of agreement only with (null) pronouns, which Nevins (2011) and Weisser (2019) claim is unattested. The claim that agreement only with pronouns is unattested is most famously challenged by observations about Modern Irish (McCloskey & Hale 1984), where we appear to see verbal inflection only with pronominal subjects. However, as noted already by McCloskey and Hale, the Irish patterns are also compatible with an analysis that the apparent 'inflection' actually reflects incorporation of subject pronouns with the verb; McCloskey and Hale suggest that this incorporation-based analysis may be less stipulative than an agreement-based one. The incorporation-based analysis continues to have relevance in the Irish literature (for a recent overview, see Bennett et al. 2019: fn.10), and is similar to the account of subject resumption that I will develop for Atchan here.

Of course, should the Atchan subject forms turn out to be pronoun-only agreement, we need not appeal to cliticization in explaining apparent subject 'resumption' in Atchan as I do here. In that case, this paper's claim of multiple sources for Atchan 'resumption' would remain trivially true, in that agreement itself could be a source of apparent 'resumption' in movement dependencies.

[14] As I will discuss in §6.2.2, exponence of IPFV aspect with L-toned verbs involves a H tone that can dock on lexical DP subjects. This H-tone docking is fully phonological predictable, unlike these portmanteau forms.

These data provide evidence that S forms are phonologically and syntactically weak. However, I want to make a broader claim that Atchan subject pronouns (in canonical subject position) are *all* syntactic clitics, even when they are not S forms. To this end, in the next section, I provide additional evidence that *all* subject pronouns associate more closely with aspectual/verbal material than lexical DPs do.

6.1.2 Subject pronouns across the paradigm

In this section, I extend beyond S forms to argue that *all* subject pronouns associate more tightly with aspectual/verbal material than lexical DPs do, and that singular subject pronouns associate yet more tightly than plural subject pronouns do.

One phonological diagnostic for differences between subject pronouns and lexical DPs comes from nasalization phenomena. As first observed by Bôle-Richard (1983) and extensively discussed by Russell (2024), subject pronouns obligatorily trigger long-distance nasalization phenomena that lexical DPs do not. The most basic data is that nasal subject pronouns in Atchan (1SG, 3SG$_{\text{PROX}}$, 2PL) nasalize verbs that they occur directly adjacent to (32), while nasal lexical DPs do not (33).[15] This is shown below, with the underlyingly oral verb /ɓá/ 'come':[16]

(32) a. mɛ̃= **m**a
1SG come
'I came.'

b. ɛ̃́= **m**â
3SG$_{\text{PROX}}$.S.PFV come
'She came.'

c. ɔ̃́= **m**â
2PL.S come
'You (pl.) came.'

(33) lep$^{\text{h}}$ã **ɓ**a
person come
'A person came.' (Russell 2024: (25))

The sum of this data, pairing S-specific aspectual portmanteau forms and the broader nasalization pattern, suggests some distinction in the statuses of pronouns and lexical DPs: pronouns exhibit aspect-related distinctions that lexical DPs do not, and they can condition verbal changes that lexical DPs cannot. Note that that 1SG data reveal that this nasalization process does not depend on the presence of a distinct S form.

An extension of the nasalization pattern also provides evidence for pronoun-internal distinctions: namely, singular pronouns associate yet more tightly with aspectual and verbal material than plural pronouns do. The key observation here from Russell (2024), building on Bôle-Richard's (1983) description, is that singular pronouns trigger a long-distance kind of nasal spreading that plural pronouns do not. Namely, nasal singular pronouns cause full nasalization of auxiliary material and (pre)nasalization of the verb, while nasal 2PL.S only triggers (pre)nasalization of the auxiliary or verb that linearly follows it:

[15] As discussed by Russell, oral subject pronouns do not trigger this nasalization process.
[16] All verbal tone alternations here are phonologically predictable; see §6.6.

(34) a. ã= **mɛ̃́** **nɛ̃́** **má**
3SG$_{PROX}$.S FUT NEG come
'She will not come.'
b. ɔ̃́= **m**ꜜá lé ɓá
2PL.S FUT NEG come
'You (pl.) will not come.' (Russell 2024: (1b–1c))

Here, the underlying form of the auxiliary and verbal material is /ɓá lé ɓá/; the 3SG subject nasalizes all auxiliary material and the initial C of the verb, while the 2PL subject only nasalizes the initial C of the future auxiliary. The analysis of this long-distance nasalization phenomenon is complex (and I refer the reader to Russell 2024 for more discussion), but it suffices for our purposes to note Russell's proposal that this phenomenon is best analyzed by positing some morphosyntactic difference between singular and plural pronouns.

This distinction between singular and plural pronouns is also reinforced by an additional observation about aspect. Specifically, while progressive aspect in Atchan is ordinarily exponed by a prefix [e-] (discussed more below), all singular pronouns cannot co-occur with [e-] (Russell 2023). This illustrated with 1SG and 3SG$_{PROX}$ forms below. Meanwhile, observe in (59) that plural pronouns like 3PL do not fuse with [e-].

(35) a. mɛ̃= (*e-) pʰí ɗja
1SG PROG laugh V.PART
'I am laughing.'
b. ã= (*e-) pʰí ɗja
3SG$_{PROX}$.S PROG laugh V.PART
'She is laughing.'

(36) wo= e-pʰí ɗja
3PL PROG-laugh V.PART
'They are laughing.' (20240720_hre_jea)

This transparently suggests an additional degree of connection between singular pronouns and progressive aspect that does not occur with plural pronouns. Note that this fusion process occurs with *all* singular pronouns: 1SG lacks a distinct S form but still fuses with the progressive.

The conclusion of this section is that pronouns (both singular and plural, and regardless of the presence of distinct S forms) associate with aspectual/verbal material in ways that lexical DPs do not. Later, I will formalize this relationship between pronouns and aspect by positing that pronouns undergo morphological operations with Asp. Note also that these nasalization processes are obligatory; however, in examples throughout the paper that contrast multiple subject forms differing in nasality, I suppress nasalization marking for consistency.

6.2 Resumptive split data

In this section, I provide the core data on subject resumption in Atchan. Note that the form of subject resumption is entwined with aspect in Atchan. Accordingly, I first discuss progressive-aspect clauses in §6.2.1, then discuss clauses with tonally-exponed aspect in §6.2.2. My discussion of Atchan aspect morphology is based on Russell 2023, which itself builds on descriptions by Bôle-Richard (1983) and Dido (2018).

6.2.1 Resumptive split: Progressive aspect

Progressive aspect in Atchan is ordinarily segmentally exponed, as a prefix [e-] PROG that prefixes onto verbs and auxiliaries:

(37) a. mɔja e-p^{h}í ɗja
M. PROG-laugh V.PART
'Moya is laughing.'

b. mɔja e-ɓá p^{h}í ɗja
M. PROG-FUT laugh V.PART
'Moya will laugh.' (20240720_hre_jea)

This [e-] prefix exhibits tonal polarity (Russell 2023): it always bears the opposite tone to its host. Atchan has two underlying contrastive tones, H and L. Throughout this section, I use *p^{h}í ɗja* 'laugh' as an example H-toned verb, and *la ɲãkã* 'pray' as an example L-toned verb. Observe the tonal polarity in these examples: when the progressive prefix affixes to 'laugh', it bears a L tone (37a); when it affixes to 'pray', it bears a H tone (38).

(38) mɔja é-la ɲãkã
M. PROG-pray God
'Moya is praying.' (20240521_kou)

In singular progressive fusion—the only time when the progressive does not surface as [e-]—the tone on the singular pronominal subject is polar to that of the verb:[17]

(39) a. { ã= / *ã= e- } p^{h}í ɗja
3SG$_{\text{PROX}}$.S 3SG$_{\text{PROX}}$.S PROG laugh V.PART
'She is laughing.'

b. { ã́= / *ã= é- } la ɲãkã
3SG$_{\text{PROX}}$.S 3SG$_{\text{PROX}}$.S PROG pray God
'She is praying.' (20240720_hre_jea)

This intuitively maintains the tonal polarity associated with the progressive.

Turning now to subject extraction in progressive clauses, we see that extracted pronominal subjects in progressive-aspect clauses are obligatorily resumed by fully phi-matching pronouns. This is illustrated below:

(40) a. mɛ̃́ nɔ̃ { ã= / *∅ e- } p^{h}í ɗja
3SG$_{\text{PROX}}$ FOC 3SG$_{\text{PROX}}$.S PROG laugh V.PART
'It's her who is laughing.' (20240310_kou)

b. wo nɔ̃́ { wo= e- / *ã= / *∅ e- } p^{h}í ɗja
3PL FOC 3PL PROG 3SG$_{\text{PROX}}$.S PROG laugh V.PART
'It's them who are laughing.' (20240423_kou)

c. hɔ̃́ nɔ̃ { hɔ̃́= e- / *wo= e- / *ã= / *∅ e- } p^{h}í ɗja
2PL FOC 2PL PROG 3PL PROG 3SG$_{\text{PROX}}$.S PROG laugh V.PART
'It's you (pl.) who are laughing.' (20240423_kou)

Here, we see that resumption is obligatory, and that only full phi-matches are permitted.

In contrast, lexical DP subjects are not resumed, in both local (41) and (for at least some speakers) long-distance (42) subject extraction:

[17] Recall that, when displaying these examples with multiple subject possibilities that differ in nasality, I omit marking pronoun-triggered prenasalization on the verb.

(41) a. jípɔ nɔ̃ { ∅ e- / *ã= } pʰí ɗja
child FOC PROG 3SG$_{\text{PROX}}$.S laugh V.PART
'It's the child who is laughing.'

b. ɛ̃́mjɔ́ nɔ̃ { ∅ e- / *wo= e- } pʰí ɗja
child.PL FOC PROG 3PL PROG laugh V.PART
'It's the children who are laughing.' (20240423_kou)

(42) kati nɔ̃ mɛ̃= mú sálé { ∅ e- / *ã= } pʰí ɗja
K. FOC 1SG think COMP PROG 3SG$_{\text{PROX}}$.S laugh V.PART
'It's Katie who I think is laughing.' (20240723_kou_jea)[18]

This is, of course, the complete inverse of pronominal extraction; compare (40a) to (41a), and (40b) to (41b). This provides our core empirical generalization: in subject extraction, pronouns are resumed but lexical DPs are not.

An important hypothesis that we will entertain and rule out is that the apparent non-resumption of lexical DP subjects could possibly be phonologically-null inanimate resumption. At first glance, this hypothesis could be compelling, since in progressive-aspect clauses, inanimate subjects are in general null:

(43) ∅ e-nɛ̃́
PROG-melt
'It is melting.' (20241217_kou)

One could, therefore, analyze the data in (41) in terms of resumption with a (null) inanimate pronoun. (See Yip & Ahenkorah 2023 for discussion of inanimate subject resumptive forms in related Akan.)

As evidence against this analysis, I turn in the next subsection to Atchan's other two aspects, perfective and imperfective. Both of these aspects are exponed tonally, and subject extraction in these aspects maintains a resumption distinction between pronouns and lexical DPs. Intriguingly, we see an additional (invariant) segmental marker of extraction in lexical-DP extraction. This reflex, however, is demonstrably *not* an inanimate pronoun, enabling us in general to rule out a story based on inanimate resumption for lexical DP subjects.

6.2.2 Resumptive split: Tonal aspect

The signature of imperfective aspect in Atchan is tonal polarity (i.e., tonal contrast) between the subject and the verb, realized as a floating tone that docks onto the subject. (For this introduction, I focus on underlyingly L-tone subjects like /mɔja/ 'Moya'.) That is, if the verb is H-toned (44a), an L-toned subject surfaces with its underlying L tone in the imperfective. By contrast, if the verb is L-toned (44b), the final vowel of the subject takes on a H tone in the imperfective. This is is illustrated below:

(44) Imperfective aspect
a. mɔ**ja** pʰí ɗja áká kʰúmbrɛ̃
M. laugh V.PART day all
'Moya laughs every day.'

[18] My data suggests some interspeaker variation on this judgment: namely, one younger speaker accepts *ã*-resumption here, while two older speakers (whose judgments are the the ones mainly reported in this paper) systematically reject it.

b. mɔjá la ɲãkã áká kʰúmbrɛ̃
M. pray God day all
'Moya prays every day.' (20240521_kou)

In (44b), the final vowel of /mɔja/ is raised to H, exponing the imperfective.

With L-toned verbs, the imperfective's H tone always docks onto the final vowel of the subject DP. This docking process is insensitive to the internal syntactic structure of the subject: for instance, it can dock to lexical N's like 'bird' in (45a), underlyingly L-toned /kɔcɛ̃/, and adjectives like 'black' in (45b), underlyingly L-toned /mru/.

(45) Imperfective aspect

a. kɔcɛ̃́ fe áká kʰúmbrɛ̃
bird fly day all
'The bird flies every day.'

b. kɔcɛ̃ mrú fe áká kʰúmbrɛ̃
bird black fly day all
'The black bird flies every day.' (20240805_kou)

I conclude that this H-tone docking onto the subject is purely phonological and derivationally late.

Meanwhile, the signature of perfective aspect is verb tone lowering. If the verb is underlyingly H-toned, the verb's tone is lowered to surface as L. Underlyingly L-toned verbs also surface as L in the perfective (i.e., tones cannot lower beyond L). This is illustrated below:

(46) Perfective aspect

a. mɔja pʰ**i** ɗja ɛ̃mpi
M. laugh V.PART one.day.away
'Moya laughed yesterday.'

b. mɔja l**a** ɲãkã ɛ̃mpi
M. pray God one.day.away
'Moya prayed yesterday.' (20240521_kou)

In (46a), underlyingly-H /pʰí/ lowers to [pʰi]. In (46b), underlyingly-L /la/ also surfaces with an L tone.

In subject extraction in these other aspects, just as in the progressive, pronouns are resumed phi-faithfully. This is shown for imperfective aspect in (47) and for perfective aspect in (48):

(47) Imperfective aspect

a. mɛ̃́ nɔ̃́ { ã= / *á= / *a / *∅ } pʰí ɗja áká kʰúmbrɛ̃
3SG$_{\text{PROX}}$ FOC 3SG$_{\text{PROX}}$.S 3INAN.S ASP laugh V.PART day all
'It's her who laughs every day.'

b. wo nɔ̃́ { wo= / *ã= / *á= / *a / *∅ } pʰí ɗja áká kʰúmbrɛ̃
3PL FOC 3PL 3SG$_{\text{PROX}}$.S 3INAN.S ASP laugh V.PART day all
'It's them who laugh every day.'

c. hɔ̃́ nɔ̃́ { ɔ̃́= / *wo= / *ã= / *á= / *a / *∅ } pʰí ɗja
2PL FOC 2PL.S 3PL 3SG_{PROX}.S 3INAN.S ASP laugh V.PART
áká kʰúmbrɛ̃
day all
'It's you (pl.) who laugh every day.' (20240805_kou)

(48) Perfective aspect[19]

a. mɛ̃́ nɔ̃́ { ɛ̃́= / *á= / *a / *∅ } pʰi ɗja
3SG_{PROX} FOC 3SG_{PROX}.S.PFV 3INAN.S ASP laugh V.PART
ɛ̃mpi
one.day.away
'It's her who laughed yesterday.'

b. wo nɔ̃́ { wo= / *ɛ̃́= / *á= / *a / *∅ } pʰi ɗja
3PL FOC 3PL 3SG_{PROX}.S.PFV 3INAN.S ASP laugh V.PART
ɛ̃mpi
one.day.away
'It's them who laughed yesterday.'

c. hɔ̃́ nɔ̃́ { ɔ̃́= / *wo= / *ɛ̃́= / *á= / *a / *∅ } pʰi
2PL FOC 2PL.S 3PL 3SG_{PROX}.S.PFV 3INAN.S ASP laugh
ɗja ɛ̃mpi
V.PART one.day.away
'It's you (pl.) who laughed yesterday.' (20240805_kou)

Interestingly, a segmental reflex, [a], arises in subject lexical-DP extraction in these two aspects. The tone of this [a] relates to the clause's aspect. In imperfective aspect, the tone of [a] is always polar to the verb:

(49) Imperfective aspect

a. mɔj**a** nɔ̃́ **á** la ɲãkã áká kʰúmbrɛ̃
M. FOC ASP pray God day all
'It's Moya who prays every day.'

b. mɔj**a** nɔ̃́ **a** pʰí ɗja áká kʰúmbrɛ̃
M. FOC ASP laugh V.PART day all
'It's Moya who laughs every day.' (20240727_kou)

Here, in contrast to (44), the subject surfaces with its underlying L tone across the board. With the L-toned verb /la/, aspect's floating tone is not lost: H-toned [á] in (49a) reflects the polar H tone of the imperfective. By contrast, in (49b), low-toned [a] is polar to H-toned /pʰí/.

Meanwhile, in the perfective, [a] is invariably L-toned, no matter the underlying tone of the verb:

(50) Perfective aspect

a. mɔja nɔ̃́ **a** la ɲãkã ɛ̃mpi
M. FOC ASP pray God one.day.away
'It's Moya who prayed yesterday.'

[19] In perfective aspect, as discussed below, H-toned subjects spread their tone onto the verb. For simplicity of presentation, since the resumptive options contrasted here differ in tone, I do not illustrate that tone spreading in these examples.

b. mɔja nɔ̃́ **a** pʰi ʤa ɛ̃mpi
M. FOC ASP laugh V.PART one.day.away
'It's Moya who laughed yesterday.' (20240727_kou)

Note that the verb also surfaces as L-toned across the board here, as is standard for perfective aspect.

I will present a positive suggestion for the status of [a] in §6.6. In this section, I want merely to note that this [a] is *not* Atchan's inanimate pronoun. In tonally-exponed aspects, the 3INAN.S pronoun is H-toned [á], while the extraction marker [a] is L-toned (which takes on H sometimes for tonal contrast in the imperfective). That the two markers are not the same can be clearly shown in perfective aspect:

(51) Perfective aspect
a. **á**= nɛ̃́ ɛ̃mpi
3INAN.S melt one.day.away
'It melted yesterday.'
b. ɛ̃ŋgwɛ̃́$_i$ nɔ̃ ∅$_i$ **a** nɛ̃ ɛ̃mpi
shea.butter FOC ASP melt one.day.away
'It's the shea butter that melted yesterday.' (20241217_kou)

Here, we see the H-toned 3INAN.S pronoun in (51a); note that perfective aspect does not alter the tone on this pronoun.[20] In contrast, the extraction marker is L-toned in (51b). This tonal distinction provides evidence that the two markers are not one and the same (contra Jarvis 2025d). More broadly, we can observe that the extraction marker [a] is not a pronoun that occurs elsewhere in Atchan; if the extraction marker were a resumptive pronoun, it would violate McCloskey's (2002) generalization that resumptive pronouns in a language are always also ordinary pronouns. Instead, I conclude (contra Dido 2018; cf. also Dido 2023) that the extraction marker [a] is not a true resumptive pronoun, and will return to this in §6.6. With this, I maintain the generalization that extracted subject pronouns in Atchan are phi-faithfully resumed, while extracted subject lexical DPs are *not* resumed by resumptive pronouns.

In the next two subsections, I turn to the analysis of this subject resumptive split. In §6.3 and §6.4 I argue that analyses based in, respectively, P-requirements and stranding cannot naturally be extended to account for the subject data, though they could each capture the PP subextraction patterns. Instead, I argue in §6.5 that a cliticization account can better capture this data.

6.3 Against a P-requirement approach

In this section, I argue that it is not possible to develop a non-stipulative P-requirement-based analysis of the subject resumption facts. The largest challenge of all comes from the fact that the reflex of lexical-DP subject movement is not a pronoun that otherwise exists in Atchan's pronominal inventory; it is fundamentally not clear how we could apply Partial Copy Deletion to yield a form that does surface in non-extraction contexts. Even setting this aside, this kind of analysis would encounter multiple other issues: empirically, we cannot use Partial Copy Deletion to derive *both* the PP and subject patterns, and the Atchan data additionally break all existing proposals of the Partial Copy Deletion algorithm across languages.

[20] As will be discussed in §6.6.2, the H-toned pronoun here spreads its tone rightward onto the perfective-marked verb.

Beginning with the final, theory-internal issue, I note that no existing formalization of the Partial Copy Deletion algorithm in the literature can derive a pattern on which pronouns are resumed but lexical DPs are not. In general, Partial Copy Deletion-based analyses are designed to capture *similarities* in resumption between pronouns and lexical DPs. The only existing account that is designed to capture resumption differences between lexical DPs and pronouns is the account of Georgi & Amaechi (2023). Unfortunately for us, their account runs in the wrong direction: the account that they develop explicitly predicts that lexical DPs must be resumed with at least as much phi-faithfulness as pronouns.[21] That is, it should be impossible to find a language that resumes pronouns with phi-faithful pronouns but does not resume lexical DPs with phi-faithful pronouns. Of course, this directly contradicts the observed Atchan facts.

Should we, then, develop a new version of the Partial Copy Deletion algorithm that can capture this pronoun/lexical DP split? I argue that we should not. The first reason for this is that I highly doubt that such an account can be successful. This is because I have already argued that we *can* appeal to Partial Copy Deletion to capture the PP subextraction data. On such an account, as discussed above, we need a version of Partial Copy Deletion that can generate resumptive pronouns that fully phi-match their antecedents (whether lexical DPs or pronouns). If we take the Partial Copy Deletion algorithm to be a language-wide algorithm (as Georgi & Amaechi (2023) explicitly do), we can only use it to derive one degree of phi-mismatch in resumptive pronouns, across the entire language. We cannot use it to do minimal tree-trimming for PP subextraction and also to trim lexical DP subjects into oblivion. Taking a language-wide perspective, it seems that the best we could do would be to posit *two* Partial Copy Deletion algorithms for Atchan: one that applies in PP subextraction, and a second that applies in subject extraction. While possibly implementable from a technical perspective, this move would make the analysis almost entirely stipulative—it seems undesirable to 'permit' post-syntactic algorithms to make explicit reference to specific syntactic positions.

Most fundamentally, Partial Copy Deletion definitionally shaves down lower movement copies to create resumptive pronouns; it is thus not clear how it could derive a pattern of selective resumption of pronouns but not lexical DPs. To maintain a subject P-requirement analysis, we could consider developing a system in which insertion of a resumptive pronoun fails for lexical DP subjects. Such a system, however, encounters its own issues.

One such approach, suggested by a reviewer, would involve positing that subject pronouns (resumptive or not) realize spans of nominal heads, rather than particular heads in the nominal structure; this is implicit in the Scott (2021)-style Vocabulary Items presented earlier. To derive a pronoun-only subject resumption pattern, we could hypothesize that subject pronouns (and not subject lexical DPs) undergo some operation that modifies their structure; for concreteness, let us say that the relevant operation is head movement of one head A in the nominal structure to another head B. Atchan subject pronoun forms, then, realize the span [D A + B]. Since, by hypothesis, lexical DPs do not undergo the A + B head movement, the subject pronoun forms cannot realize the span [D A B], hence (we hope) no resumption of subject lexical DPs.

While this approach might derive the absence of *subject* pronoun forms in lexical-DP subject resumption, as far as I can tell it does not predict the total absence of resumption in subject position. Instead, this approach would predict that the most-specified form that can realize the [D A B] span should be inserted to resume a moved lexical DP. It seems that the best candidate to realize this span would be an elsewhere pronoun. We

[21] See '$?_2$' and '$?_3$' in Georgi & Amaechi's (2023) table 4, and related discussion on p.999.

know from PP subextraction that elsewhere pronoun forms can resume both non-subject pronouns and non-subject lexical DPs; furthermore, the data in (28) shows that these forms can occur in various syntactic positions and likely should not be thought of as exponents of a specific case. Accordingly, it seems that the elsewhere pronoun form should be a viable item to insert in this position. However, contrary to this position, moved subject lexical DPs cannot be resumed by elsewhere pronouns:

(52) * jípɔ nɔ̃ mɛ̃́ (m)pʰí ɗja ɛ̃mpi
child FOC 3SG$_{\text{PROX}}$ laugh V.PART one.day.away
Intended: 'It's the child who laughed yesterday.' (Maxime Dido p.c.)

To repair this, we would need to come up with an explanation for why elsewhere pronouns cannot be inserted here. Indeed, it would be necessary that *no* non-null nominal form in Atchan could be inserted in this position. At this point, it is not clear that such an approach can be pursued without a number of stipulations that, as far as I can tell, lack empirical motivation.[22]

For these reasons, I conclude that we cannot straightforwardly appeal to a subject P-requirement to derive the subject resumption data.

6.4 Against a stranding approach

An attempted stranding analysis of the subject resumption data encounter similar issues to the P-requirement approach: it is similarly unclear (though for slightly different reasons) how to derive position- and item-specificity in subject pronoun-only resumption.

In its most basic form, a stranding approach predicts that, whenever the Big DP structure is generated, we should see a fully phi-matching resumptive pronoun at the tail of the movement dependency. This property made stranding an appealing line of analysis for the PP object resumption pattern, but making the same claim about subject position as PP object position clearly predicts resumption of both lexical DP and pronominal subjects. Two ways of modifying the analysis to capture the pronoun-only resumption pattern present themselves: claiming that the additional D_2 in lexical DPs goes unrealized, or claiming that no Big DP is generated in lexical-DP subject extraction. Neither of these avenues seems promising, for reasons that I discuss below.

First, I consider the claim that D_2 is present in lexical-DP subject extraction but is not pronounced: i.e., a null element can realize a moved lexical-DP subject in this position. This amounts to a clear violation of McCloskey's (2002) generalization that all resumptive pronouns in a language are also ordinary pronouns in that language. As we saw from the aspect data, there is no generally-available null subject pronoun across aspects in Atchan. Positing this null resumptive therefore would entail positing a pronominal element that does not otherwise occur in non-extraction environments and would carry a high theoretical cost.

Second, the claim that the D_2-containing Big DP structure is not projected in lexical-DP subject extraction is also not especially appealing. To try formalizing such a claim, we might think about drawing a parallel to well-known information-structural restrictions on clitic doubling in languages like Spanish, which Hewett (2023: 187) suggests to implement by pronominal D_2 selecting for nominals bearing particular features. Assuming a feature like $[+\pi]$ that distinguishes lexical DPs and pronouns (Harley & Ritter 2002), we could posit that D_2 selects for $[+\pi]$ nominals. However, by making such a claim we

[22] Note that other post-syntactic operations like Impoverishment (cf. Ershova 2024) also do not fare well, since they similarly are intended to always derive pronominal forms.

lose the ability to explain across-the-board resumption of PP objects. Instead, we seemingly must appeal to two different varieties of D_2, depending on the syntactic position from which we are extracting. Overall, this situation exactly mirrors the difficulties with the P-requirement analysis that arose in the previous section.

Repairing the analysis is also challenging because subject position is, presumably, a derived position; we cannot entirely straightforwardly ascribe differences in D_2's selectional behavior to where it is merged. If the pronoun-only resumption occurred only with unergative and transitive predicates (like the data on 'laugh' and 'pray' observed to this point), we could plausibly say that D_2 merged in Spec,*v*P (or wherever agents are introduced) imposed its own selectional restrictions that D_2 merged inside PPs did not, perhaps enabling us to derive the distinctions. However, as it turns out, the same pronoun-only resumption pattern is observed with presumably unaccusative predicates like *k^h^u* 'die':

(53) a. jípɔ nɔ̃ { a / *ɛ̃́= } k^hu
child FOC ASP $3SG_{PROX}$.S.PFV die
'It's the child who died.'

b. mɛ̃́ nɔ̃ { ɛ̃́= / *a } k^hu
$3SG_{PROX}$ FOC $3SG_{PROX}$.S.PFV ASP die
'It's her who died.' (Maxime Dido p.c.)

On the Unaccusative Hypothesis (Perlmutter 1978; Burzio 1986), the subjects in these sentences are generated as the complement to the verb, a position that we know does not permit resumption in Atchan. We therefore cannot tie these posited differences in D_2's selectional behavior to properties of where it is base-merged.

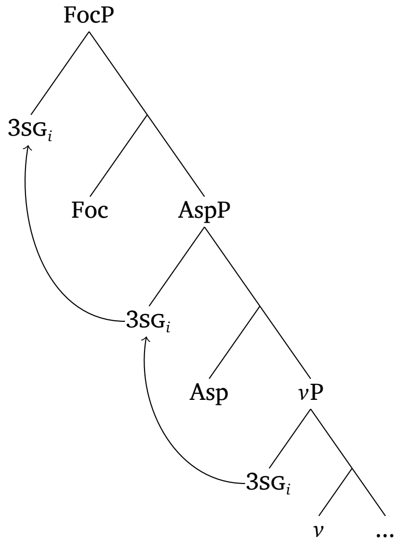
For these reasons, I conclude that a stranding-based account also fails to cleanly capture the subject resumption data.

6.5 In support of a cliticization approach

In this section, I argue cliticization is best-suited to account for the pronoun-only resumption of subjects in Atchan. The takeaway of §6.1 was that subject pronouns in Atchan are more closely entwined with aspectual/verbal material than subject lexical DPs are: only subject pronouns trigger verbal allophony and undergo aspect-based fusion and portmanteau processes. These empirical observations, I argue, lend credence to the idea that resumption of extracted pronouns is linked to subject pronoun cliticization. In this section, this close relationship between pronouns and aspectual material will be formalized using morphological operations that tie together these pronouns and the Asp head.

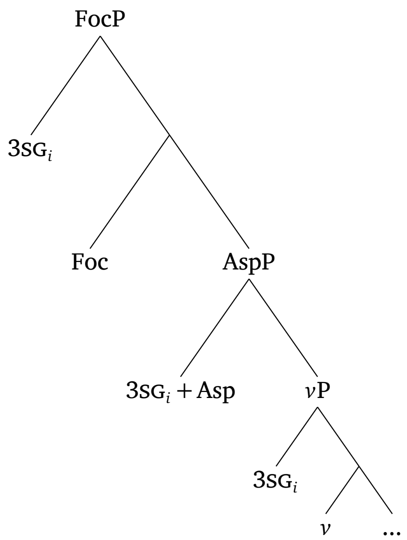
The cliticization analysis introduced in §4.2 can be extended straightforwardly to resumption of Atchan singular subject pronouns, since they intuitively fuse with aspect. (I will return to plural pronouns in the next section.) Formally, I assume that agentive subjects in Atchan are base-generated in Spec,*v*P (this is not crucial), and that all subjects in Atchan move to Spec,AspP for EPP reasons (i.e., Spec,AspP is the canonical subject position in Atchan, analogous to Spec,TP in languages like English). Accordingly, Step 1 for an Atchan 3SG pronominal subject that undergoes later focus movement is schematized below (with only relevant projections shown):

(54) **Step 1: Movement of 3SG$_{\text{PROX}}$ pronoun**



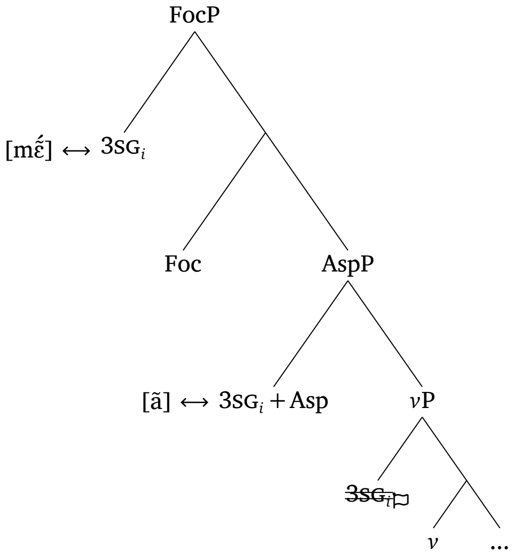
In Step 2, the singular pronoun occupying Spec,AspP undergoes Fusion with Asp:

(55) **Step 2: Fusion of Spec,AspP copy with Asp**



In Step 3, the Fused copy is invisible to Chain Reduction, so only the Spec,vP copy is tagged for non-pronunciation. This leads to double spellout of the singular pronoun in Step 4:

(56) **Step 3 + 4: Chain Reduction tagging and Vocabulary Instertion**

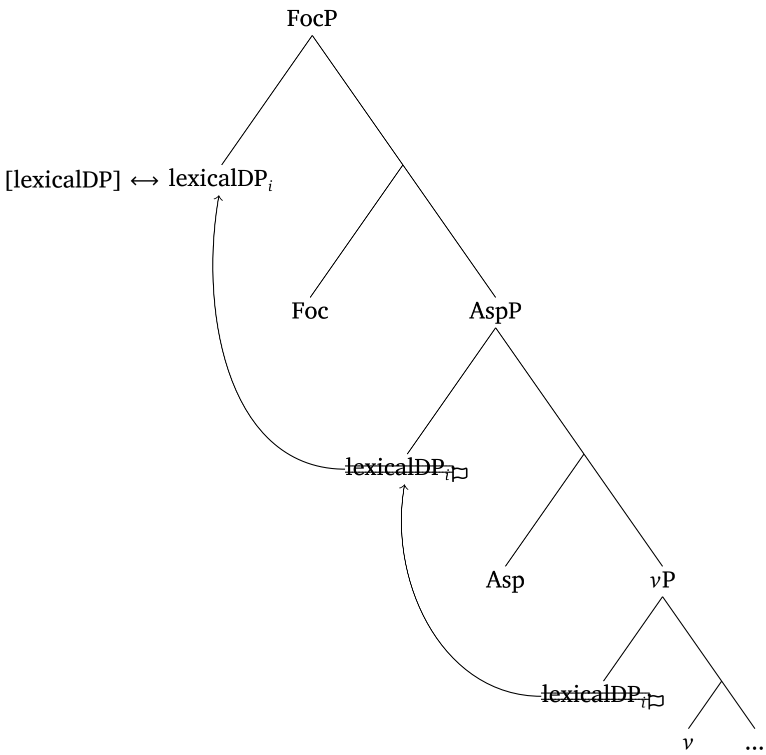


This analysis captures obligatory pronominal resumption of extracted singular subjects, as shown in (40a) and repeated below:

(57) mɛ̃́ nɔ̃ { ã= / *∅ e- } pʰí ɗja
3SG$_{\text{PROX}}$ FOC 3SG$_{\text{PROX}}$.S PROG laugh V.PART
'It's her who is laughing.' (20240310_kou)

To explain non-resumption of lexical DP subjects, I assume that they move through Spec,AspP in Step 1, for EPP reasons, but do not undergo Fusion with Asp in Step 2. This means that both lower copies are tagged for non-pronunciation, and only the Spec,FocP copy is pronounced. This entire derivation is summarized in the following tree:

(58)



In this way, a cliticization analysis is able to naturally capture the split in resumption morphology between pronouns and lexical DPs. This suggests that we should appeal to multiple different mechanisms in analyzing resumptive pronouns across languages (and within a single language). While P-requirements and stranding are well-prepared to capture *uniformity* of resumption in extraction from a given syntactic position (like the complement of P), I contend that a cliticization approach is better equipped to capture resumption of only some elements (like just pronouns, in contrast to lexical DPs).

6.6 Tying up loose ends

In this section, I address two analytical loose ends and questions for future work.

6.6.1 Resumption of plural pronouns, and m-merger

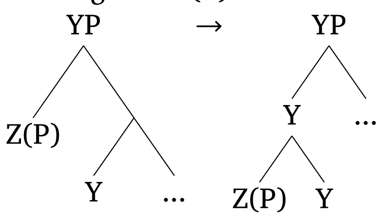
The first concerns resumption of plural pronouns in Atchan. In the previous section, I proposed a link between resumption and Fusion of singular pronouns; this Fusion is analytically justified on the grounds that singular pronouns and the segmental progressive [e-] cannot co-occur. In contrast, we saw in (36), repeated below, that plural pronouns can co-occur with the segmental progressive:

(59) wo= e-p^{h}í d͡ja
3PL PROG-laugh V.PART
'They are laughing.' (20240720_hre_jea)

This data suggests that we should *not* claim that plural pronouns undergo Fusion with Asp.

How, then, should we explain the fact that even plural extracted subject pronouns are resumed? Recall from §6.1 that, despite the absence of plural progressive fusion, there is still evidence for an intermediate degree of connectedness between plural pronouns and aspectual/verbal material. We could assimilate plural pronouns to this general resumption story by following Harizanov (2014) and Kramer (2014) in assuming that the DM operation of m-merger (Matushansky 2006) is sufficient to cause Chain Reduction invisibility. In the m-merger operation, an element Z(P) in the specifier of YP is lowered to form a complex head with Y, as schematized below:

(60) M-merger of Z(P) to Y:



The crucial difference between m-merger and Fusion is that, in the former, the two items remain distinct within the complex head. This allows for the insertion of two different lexical items to realize Z(P) and Y, unlike the obligatory portmanteau insertion with Fused elements.

To capture the intermediate degree of connectedness between plural pronouns and aspectual/verbal material, we could propose that they undergo only m-merger with Asp; note that Russell (2024) has proposed a roughly similar analysis of Atchan subject pronouns based on other data. Harizanov (2014) and Kramer (2014) propose that m-merger alone can cause Chain Reduction invisibility and consequent resumption, to capture clitic doubling in Bulgarian and Amharic, respectively. If we adopt the same view for Atchan, we can explain why *all* Atchan subject pronouns are resumed.

Note that an appeal to m-merger raises at least two larger theoretical questions, which must be left to future work. The first concerns why m-merger suffices for Chain Reduction invisibility. Harizanov (2014: 1072) suggests for Bulgarian that derived complex heads are simply generally invisible for Chain Reduction. This story, however, raises more general questions about opaqueness in head movement, and more work is warranted.

A second question raised by this analysis concerns empirical differences between the resumptive patterns addressed in this paper and the clitic doubling patterns addressed by Harizanov and Kramer. The clitic doubling phenomena that they investigate involve a movement chain whose *highest* copy undergoes m-merger and becomes opaque to the Chain Reduction algorithm, resulting in spellout of both the m-merged copy and the second-highest copy in the movement chain. In contrast, this paper has exclusively focused on data in which the m-merging copy is not the highest copy. Expanding the data under consideration, we might expect to see similar clitic doubling-like effects if a pronoun's movement terminated in subject position. However, as shown below with an unaccusative predicate whose subject presumably moves from a lower position, only one copy of the pronoun is realized:[23]

(61) ɛ̃́= { khu / *k^{h}wi }
3SG$_{PROX}$.S.PFV die die.3SG$_{PROX}$.O
'He died.' (20240521_kou)

Clearly, we must appeal to some cross-linguistic difference to explain the contrast between Atchan and Amharic/Bulgarian here.[24] While a fuller investigation must be left to future work, it seems potentially promising to attribute this difference to a contrast in the status of highest movement copies: in Amharic/Bulgarian, highest movement copies become invisible to Chain Reduction through m-merger, but in Atchan, they remain visible for the purposes of determining the movement chain's highest copy.

6.6.2 Tonally-exponed aspect and recoverability

The second loose end to tie up concerns the extraction marker [a], which we have seen surfaces in subject lexical DP extraction in imperfective and perfective aspects. The crucial data on lexical DP extraction in these aspects is repeated below, from (49)-(50):

(62) Imperfective aspect
a. mɔja nɔ̃́ **á** la ɲãkã áká k^{h}úmbrɛ̃
M. FOC ASP pray God day all
'It's Moya who prays every day.'
b. mɔja nɔ̃́ **a** p^{h}í ɗja áká k^{h}úmbrɛ̃
M. FOC ASP laugh V.PART day all
'It's Moya who laughs every day.' (20240727_kou)

(63) Perfective aspect
a. mɔja nɔ̃́ **a** la ɲãkã ɛ̃mpi
M. FOC ASP pray God one.day.away
'It's Moya who prayed yesterday.'

[23] The underlying form of the second, ungrammatical option is /k^{h}u-ɛ/ 'die-3SG.O'.
[24] In fact, since this example involves a singular pronoun (which I have assumed undergoes Fusion), the question is a larger one about Atchan movement chains, not one about m-merger specifically.

b. mɔja nɔ̃́ **a** pʰi ɗja ɛ̃mpi
M. FOC ASP laugh V.PART one.day.away
'It's Moya who laughed yesterday.' (20240727_kou)

The analysis sketched in the previous section can capture total non-resumption in progressive-aspect clauses, but the status of [a] in (62)-(63) remains unresolved. What I suggest here is that [a], at least at some level, fulfills a two-part requirement that Atchan aspect must be (i) realized (i.e., floating tones must have sufficiently-local hosts), in (ii) a fully recoverable way. I suggest that [a] is a last-resort form (roughly epenthetic, analogously to Rolle & Merrill 2022 on tone-driven epenthesis) inserted when these two requirements are not otherwise satisfiable.

The first part of the requirement explains the appearance of [a] in imperfective-marked clauses. Atchan imperfective can be analyzed as a floating tone polar to the verb (Russell 2023), which needs a host in order to satisfy (i). Presumably, FOC is insufficiently close to host the floating tone, so [a] is inserted and bears that tone.

More must be said for the perfective, since (unlike the imperective) the perfective's tonal lowering on the verb is 'successful' even in cases of extraction: observe the L tone on [pʰi] 'laugh' in (63b). I suggest that [a] surfaces in perfective-aspect extraction because perfective exponence is phonologically messy. In particular, as illustrated below, H-toned subjects spread and overwrite the L tone that the perfective introduces (Russell 2023):

(64) Perfective aspect

a. mɔja pʰ**i** ɗja ɛ̃mpi
M. laugh V.PART one.day.away
'Moya laughed yesterday.' (20240521_kou)

b. jajó pʰ**í** ɗja ɛ̃mpi
Y. laugh V.PART one.day.away
'Yayo laughed yesterday.' (20240727_kou)

Here, we see two perfective-aspect clauses, which differ underlyingly only in the tone of the subject's final vowel. Both involve the underlyingly H-toned verb /pʰí/ 'laugh', whose tone is replaced with L in the perfective. In (64a), /mɔja/ is underlyingly L, while in (64b) the final vowel in /jajó/ is underlyingly H. In perfective clauses with H-toned subjects, the H tone spreads rightward onto the perfective-marked L-toned verb, yielding the surface H-toned [pʰí] in (64b). The effect is a kind of Duke-of-York derivation: the perfective lowers the verbal tone from /H/ to L, only for tone spreading to raise it back to surface [H] in (64b). In summary, to unambiguously know the aspect of an Atchan clause, we cannot look just at the verb's surface tone: we must also know the subject's tone, to determine if tone spreading has occurred. I suggest that extraction [a] is fulfilling this need to know the immediately preverbal tone, so that aspect is fully recoverable.

Note that this requirement to have a preverbal tone can be formalized without appealing to recoverability *per se*: for example, we could propose a linearity requirement: the verb/auxiliary must not be the first overt item within the AspP domain. A fuller characterization of the extraction marker [a] and why it is the best option to insert in this environment, however, must be left to future work.

7 Conclusion

This paper has investigated the morphosyntax of resumption in Atchan, with an eye towards the algorithms in the grammar that permit and block resumption. I have demonstrated that Atchan exhibits multiple splits in movement-based resumption: in some positions (namely, PP subextraction) resumption is obligatory across all DP types, while in other positions (namely, subject extraction) resumption is obligatory for pronouns but banned for lexical DPs. The details of these splits, I have argued, require us to recognize multiple mechanisms in the grammar that can give rise to resumption.

The overall picture that I have argued for is one on which non-pronunciation of multiple movement chain copies—i.e., chain reduction—is due to economy. However, multiple different mechanisms can override these constraints or make them inapplicable. In some cases, idiosyncratic morphological processes shift elements around in the structure, moving them into domains in which they are shielded from the economy algorithm. In other positions, particular (language-specific) positions can be associated with prosodic prominence and require overt pronunciation, or a more complex external syntax of lower movement copies can give rise to resumption.

This appeal to multiple resumptive mechanisms in Atchan raises the question of how analytical weight is distributed across patterns and languages. One consequence of the argument in this paper is the identification of item-specific resumption as, perhaps, cross-linguistically suggestive of cliticization at play. In particular, the analysis of subject resumption that I have developed here relies on a proposal that all and only pronouns in Atchan undergo key morphological cliticization operations. It is conceivable, however, that different sets of elements might undergo the relevant operations in other languages. This restricted scope might serve as a cross-linguistically relevant diagnostic, since P-requirements would in general be predicted to apply to all elements occupying a particular position (barring language-specific null morphemes, cf. van Urk 2018: fn.22). See also Hewett 2023: 185,194 for similar discussion of how item-specificity might be captured in a stranding framework.

A second prediction of this view, which must await further cross-linguistic testing, is that the two different mechanisms proposed here predict different ranges of possible phi-mismatches in resumption across languages. In particular, the P-requirement and stranding approaches that I have discussed here can be parametrized to account for phi-mismatches in resumption in different languages (cf. especially Georgi & Amaechi 2023 on P-requirements and Hewett 2023: 474–475 for a relevant suggestion for stranding). In contrast, the cliticization-based story is not as flexible in dealing with phi-mismatches, since operations like Fusion do not standardly delete phi-features. We therefore might expect that cliticization-based resumption should exhibit fewer phi-mismatches across languages. Languages that exhibit different degrees of phi-faithfulness in resumption will be crucial in exploring the accuracy of this prediction.

Abbreviations

ASP = aspect, COMP = complementizer, COP = copula, FOC = marker, FUT = future, INAN = inanimate, IPFV imperfective, NEG = negation, PFV = perfective, PL = plural, PROG = progressive, PROX = proximate, O = object, OBV = obviative, S = subject, SG = singular, TOP = topic marker, V.PART verbal particle in a particle-verb construction.

Ethics and consent

The Atchan documentation project was approved by the UC Berkeley Committee for Protection of Human Subjects, Protocol ID 2021-06-14420. All Atchan consultants gave informed consent to participate in this research.

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Competing interests

The author has no competing interests to declare.

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